

Transformer Heading Estimation: Model Improvement Report

Results Comparison (Before v/s After)

Overall Performance

The table below summarises the MAE and RMSE across all four training runs, from the original baseline through to the final improved model.

Run	Changes Made	MAE (°)	RMSE (°)
Original (baseline)	No changes	66.99	83.30
Run 2	feedforward 512 + lr 0.0001 + window 400	67.04	83.20
Run 3	dropout 0.3 only	66.72	82.98
Run 4 (best)	dropout 0.3 + window 400 + stride 100	66.65	82.44

The most significant change happened in Run 4 when the dropout of 0.3 was applied alongside a bigger time window stride of 400 timesteps to the network. This combination gave the best MAE value of 66.65° and RMSE value of 82.44°, showing a decrease in 0.34° for MAE and 0.86° for RMSE compared to the original one. What is more important, is that there was no increase in performance from Run 2 where the feedforward of 512 was used, along with the learning rate of 0.0001, and the time window of 400.

Scenario Breakdown

The table below compares per-scenario MAE between the original model and the best improved model (Run 4).

Scenario	Original MAE (°)	Improved MAE (°)	Windows
Straight walking	60.57	59.06	10,807
Sharp turns	86.05	82.19	10,731
Short trajectories	68.09	69.14	19,196
Long trajectories	66.08	64.17	19,196

The largest improvements made per scenario was in sharp turns, where MAE improved from 86.05° to 82.19°, making a difference of 3.86°. This aligns with the proposed assumption that using an even larger

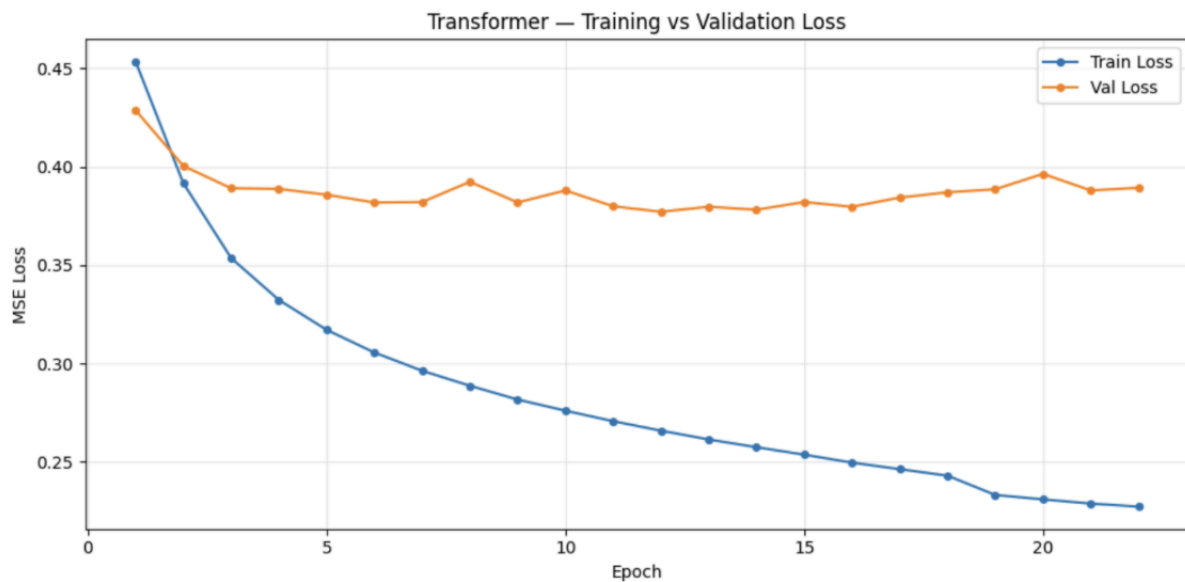
window of 400 timesteps provides enough context for detecting direction change over one second. The straight walking scenario improved from 60.57° to 59.06° . Long trajectory scenarios had MAE improve from 66.08° to 64.17° , while the short trajectories showed only minor improvements from 68.09° to 69.14° .

Quantitative Results (Improved Model)

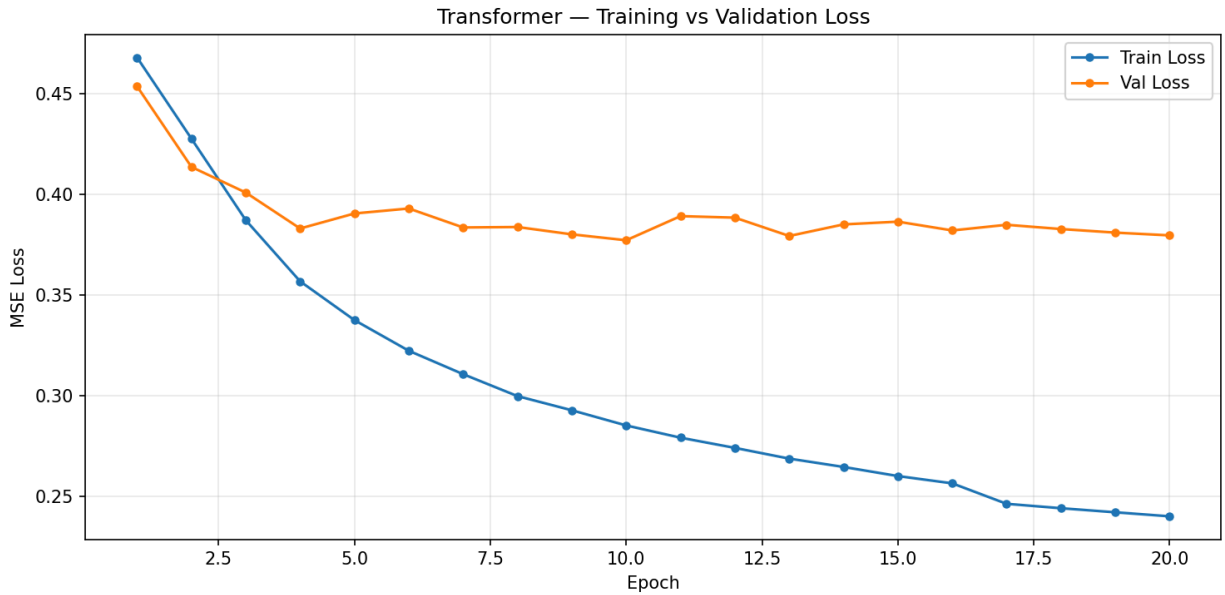
Metric	Value
MAE on test set (degrees)	66.65°
RMSE on test set (degrees)	82.44°
Approximate training time	~1 hour 20 minutes

Loss Curves

Original model loss curve (early stopped at epoch 16, best validation loss: 0.379):



Improved model loss curve (early stopped at epoch 20, best validation loss: 0.377):



Reflection Part-I

Q - 1

The Transformer failed to distinguish itself from the LSTM despite making the application adjustments. The original Transformer had an MAE value of 66.99° while the LSTM was just slightly higher at 67.05° ; the improved Transformer was only slightly better, at 66.65° . There was no noticeable difference between the two models.

This tells us something important about the nature of the task. Self-attention's core advantage is its ability to model long-range dependencies across an entire sequence in parallel, while recurrent gating processes the sequence step by step. For a task where the relevant patterns are largely local — short bursts of acceleration and rotation within a 1-2 second window — the architectural difference does not translate into a performance difference. Both mechanisms appear to reach the same performance ceiling, which is primarily determined by the dataset rather than the model

This means that when it comes to using this particular dataset for the problem of inertial heading prediction, whether to choose self-attention or recurrent gating does not matter much compared to the problem of generalization from novel subjects. Neither approach has any inherent superiority when dealing with subject-specificity becomes the bottleneck.

Q - 2

The learning rate reduction to 0.0001 (Run 2) did not improve the problem of stagnation. In the first case, the initial model achieved a performance plateau starting from epoch 6 with a minimum validation loss of 0.379. In Run 2, the model also showed similar performance, which plateaued from epoch 22 with a minimum validation loss of 0.377.

If the validation loss is stuck while the training loss keeps on decreasing then it's a sure sign of overfitting and not that the learning rate is too big. When the learning rate is reduced, this will cause the optimiser to move slower and thus helps in preventing the optimiser from overstepping the minimum. However, this can only happen when there isn't an overfit problem.

Considering the consistent plateau regardless of the difference in learning rates, the issue doesn't lie with the optimizer. The validation error plateaus since there's nothing more that the model can learn from the training subjects. All that training only reinforces the fact that the model is very adept at learning their style of movement.

Q - 3

Yes. The improvement in sharp turn was MAE from 86.05° to 82.19° , showing the most significant improvement in one scenario among all. The hypothesis that large windows help attention mechanisms detect directional changes during sharp turns can be validated by this result.

It is because of the structure of sharp turns. There is a heading shift in which there are changes within a period of time. Within 200 timesteps (which is equivalent to 1 second), the model might not be able to fully recognise a complete sharp turn, but it could only detect the beginning or ending of the turn. Therefore, a window of 400 timesteps (2 seconds) is more effective for detecting directional changes.

Self-attention takes advantage of this fact in particular since it can pay attention to both these positions at once in a single shot irrespective of the distance between them. In contrast to an LSTM where the input turn has to be processed in sequence and where the data has to be moved to the subsequent hidden state, here the turn at its initial and final positions can be compared directly by the Transformer. It is particularly advantageous when the complete turn is available inside the window due to the increased window size of 400 steps.

However, the improvement is significant but not revolutionary, since in the case of sharp turns, the angle of 82.19° still remains the most difficult to predict.

Q - 4

From the findings, it is clear that the main bottleneck is the data, particularly that the test set is made up of all unseen subjects. These findings can be justified by noting that:

- There was no improvement when the architecture was varied (feedforward dimensions and number of layers), thus ruling out model capacity as the bottleneck.
- There was no improvement when the learning rate was varied, ruling out optimization as the bottleneck.
- The difference between train and validation loss remained constant despite any change to the architecture.
- There was always 0.22-0.24 train loss vs 0.37-0.38 validation loss at stopping time.

The effect of window size was a less critical factor. The 400 step window led to a gain of 3.86° , an important but still insufficient gain in accuracy to make sharp turns the least accurate of all scenarios. Window size assists but is not a solution to the generalisation problem.

The additional dropout rate (up from 0.1 to 0.3) had a minor yet measurable impact, proving overfitting was one of the key contributing factors. The improvement was however rather minimal (0.34° MAE), indicating that with better generalisation techniques, the inherent variability of the IMU sensors remains problematic.

If provided additional time, improvements that would make the biggest difference in terms of enhancing performance would include: (1) data augmentation using random rotations of heading angles along with adding Gaussian noise to sensors; (2) domain adversarial learning, in which case the model will be trained to output heading predictions independent of the subject identity; and (3) utilization of the CLS token mechanism, where instead of taking the average pool of all time steps, the model will be able to determine those timesteps that provide more information about the heading prediction.

Reflection Part - II